Dummit & Foote Notes

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Contents

0 Preliminaries

0.1 Basics

◆ Order/Cardinality: denoted as $|A|$ for a set A. If A is finite, then $|A|$ is the number of elements in A.
◆ Cartesian Product: The Cartesian product of two sets A and B is the collection of elements

$$A \times B = \{(a, b) \mid a \in A, b \in B\}.$$

◆ Function: Denoted as $f : A \longrightarrow B$ for sets A and B. The value of f at a is denoted as $f(a)$ for all $a \in A$. Moreover, to denote $f(a) = b$, we write $f : a \mapsto b$, or $a \mapsto b$.
◆ Domain: The set A in the function $f : A \longrightarrow B$.
◆ Codomain: The set B in the function $f : A \longrightarrow B$.
◆ Range/Image: The subset of B defined as

$$f(A) = \{b \in B \mid f(a) = b \text{ for some } a \in A\}.$$

◆ Preimage/Inverse Image: For $C \subseteq B$, the subset of A given by

$$f^{-1}(C) = \{a \in A \mid f(a) \in C\}.$$

◆ Fiber: For $b \in B$, the preimage of b, i.e., $f^{-1}(b)$.

◆ Identity Map: The function on a set A defined as $\mathrm{id}_A : A \longrightarrow A$ where $\mathrm{id}_A(a) = a$ for all $a \in A$.
◆ Composition Map: Given functions $f : A \longrightarrow B$ and $g : B \longrightarrow C$, it is the map $g \circ f : A \longrightarrow C$ is defined as

$$(g \circ f)(a) = g(f(a)) \quad \text{for all } a \in A.$$

Let $f : A \longrightarrow B$ be a function.
◆ Injective/Injection: for all $a_1, a_2 \in A$, $a_1 \neq a_2$ implies $f(a_1) \neq f(a_2)$.
◆ Surjective/Surjection: for all $b \in B$, there exists $a \in A$ such that $f(a) = b$, i.e., $f(A) = B$.
◆ Bijective/Bijection: f is both injective and surjective.
◆ Left Inverse: A function $g : B \longrightarrow A$ is a left inverse of $f : A \longrightarrow B$ if $g \circ f = \mathrm{id}_A$.
◆ Right Inverse: A function $h : B \longrightarrow A$ is a right inverse of $f : A \longrightarrow B$ if $f \circ h = \mathrm{id}_B$.

♣ Proposition 0.1 ➤ Characterizations of Injectivity, Surjectivity, and Bijectivity

Let $f : A \longrightarrow B$ be a function.
(1) f is injective if and only if f has a left inverse.
(2) f is surjective if and only if f has a right inverse.
(3) f is a bijection if and only if there exists $g : B \longrightarrow A$ such that $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$.
(4) If A and B are finite sets with $|A| = |B|$, then the following are equivalent:
  (a) f is bijective.
  (b) f is injective.
  (c) f is surjective.

Proof.
(1) ➡ Suppose f has a left inverse $g : B \longrightarrow A$. Then for all $a_1, a_2 \in A$, if $f(a_1) = f(a_2)$, then

$$a_1 = g(f(a_1)) = g(f(a_2)) = a_2,$$

so f is injective.

⬅ Suppose f is injective. Then for each $b \in f(A)$, there exists a unique $a \in A$ such that $f(a) = b$. Now, fix some $a_0 \in A$. Then we can define a function $g : B \to A$ as

$$g(b) = \begin{cases} a & \text{if } b \in f(A) \text{ and } f(a) = b, \\ a_0 & \text{if } b \in B \setminus f(A). \end{cases}$$

To see g is well-defined, note that if $b \in f(A)$, then by injectivity of f, there is a unique $a \in A$ such that $f(a) = b$. Then for all $a \in A$,

$$g(f(a)) = a,$$

so $g \circ f = \mathrm{id}_A$ and g is a left inverse of f.

(2) ➡ Suppose f has a right inverse $h : B \to A$. Then for all $b \in B$, we may set $a = h(b) \in A$. Then $f(a) = f(h(b)) = b$, so f is surjective.
⬅ Suppose f is surjective. Then $f^{-1}(b)$ is nonempty for all $b \in B$. Define a function $h : B \to A$ by choosing some $a \in f^{-1}(b)$ for each $b \in B$, where this choice is possible by the axiom of choice. Then for all $b \in B$,

$$f(h(b)) = b,$$

so $f \circ h = \mathrm{id}_B$ and h is a right inverse of f.

(3) ➡ Suppose f is bijective. Then by (1) and (2), f has a left inverse $g : B \to A$ and a right inverse $h : B \to A$. Then for all $b \in B$,

$$g(b) = g(f(h(b))) = h(b),$$

where $g(f(h(b))) = g(b)$ because $f(h(b)) = b$ from h being a right inverse of f, and $g(f(h(b))) = h(b)$ because g is a left inverse of f. Thus, $g = h$ and $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$.
⬅ Suppose there exists $g : B \to A$ such that $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$. Then by (1) and (2), f is injective and surjective, so f is bijective.

(4)
- If f is bijective, then by (1) and (2), f is injective and surjective.
- If f is injective, then distinct elements of A map to distinct elements of B. Since $|A| = |B|$, this implies that f is surjective, so f is bijective.
- If f is surjective, then all elements of B are mapped to by elements of A. Since $|A| = |B|$, this implies that f is injective, so f is bijective. □

◆ Inverse: The function $g : B \to A$ in Proposition 0.1 (3), denoted as $f^{-1} : B \to A$.
◆ Permutation: A bijection $f : A \to A$.
◆ Restriction: Let $A \subseteq B$ and $f : B \to C$ be a function. The restriction of f to A is the function $f|_A : A \to C$ defined as

$$f|_A(a) = f(a) \quad \text{for all } a \in A.$$

◆ Extension: Let $A \subseteq B$ and $g : A \to C$ be a function. The extension of g to B is a function $f : B \to C$ such that $f|_A = g$. This function need not be unique, and may not even exist.

Let A be a set.
◆ Binary Relation: A subset $R \subseteq A \times A$, where we write $a \sim b$ if $(a, b) \in R$.
◆ Reflexive: A relation R is reflexive if $a \sim a$ for all $a \in A$.
◆ Symmetric: A relation R is symmetric if $a \sim b$ implies $b \sim a$ for all $a, b \in A$.
◆ Transitive: A relation R is transitive if $a \sim b$ and $b \sim c$ implies $a \sim c$ for all $a, b, c \in A$.
◆ Equivalence Relation: A relation R that is reflexive, symmetric, and transitive.
◆ Equivalence Class: For $a \in A$, the equivalence class of a is the set

$$[a] = \{b \in A \mid b \sim a\}.$$

Elements of the $[a]$ are said to be equivalent to a and are representatives of $[a]$.
◆ Partition: Let I be an indexing set. A partition of A is a collection of nonempty subsets $\{A_i\}_{i \in I}$ such that

(1) $A = \bigcup_{i \in I} A_i$.
(2) $A_i \cap A_j = \varnothing$ for all $i, j \in I$ with $i \neq j$.

Proposition 0.2 ➤ Fundamental Theorem of Equivalence Relations

Let A be a nonempty set.

(1) If $\sim$ is an equivalence relation on A, then the set of equivalence classes of $\sim$ form a partition of A.
(2) Let I be an indexing set. If $\{A_i \mid i \in I\}$ is a partition of A, then there is an equivalence relation on A whose equivalence classes are the sets A_i for $i \in I$.

Proof.

(1) Let $\sim$ be an equivalence relation on A, and put

$$B = \bigcup_{a \in A} [a]$$

to be the union of all equivalence classes of $\sim$. For any $a \in A$, we have $a \sim a$ by reflexivity, so $a \in [a]$ and thus $a \in B$. Hence, $A \subseteq B$. Conversely, let $b \in B$. Then $b \in [a]$ for some $a \in A$, so $b \sim a$ and thus $b \in A$. Hence, $B \subseteq A$ and thus $A = B$.

Now, let $a, b \in A$. If $[a] \cap [b] = \varnothing$, then we are done. Otherwise, let $c \in [a] \cap [b]$. Then $c \sim a$ and $c \sim b$, so by symmetry and transitivity, we have $a \sim c$ and $c \sim b$, which implies $a \sim b$. Now, let $x \in [a]$. Then $x \sim a$ and $a \sim b$, so by transitivity, we have $x \sim b$ and thus $x \in [b]$. Hence, $[a] \subseteq [b]$. A similar argument shows that $[b] \subseteq [a]$, so $[a] = [b]$, and thus the equivalence classes of $\sim$ form a partition of A.

(2) Let $\{A_i \mid i \in I\}$ be a partition of A. Define a relation $\sim$ on A by $a \sim b$ if and only if there exists $i \in I$ such that $a, b \in A_i$. Then $\sim$ is reflexive because for any $a \in A$, there exists $i \in I$ such that $a \in A_i$, so $a \sim a$. Moreover, $\sim$ is symmetric because if $a \sim b$, then there exists $i \in I$ such that $a, b \in A_i$, so $b \sim a$. Finally, $\sim$ is transitive because if $a \sim b$ and $b \sim c$, then there exist $i, j \in I$ such that $a, b \in A_i$ and $b, c \in A_j$. Since $b \in A_i \cap A_j$, we must have $i = j$ because the A_i are disjoint for $i \neq j$. Hence, $a, c \in A_i$, so $a \sim c$. Therefore, $\sim$ is an equivalence relation, and its equivalence classes are precisely the sets A_i for $i \in I$. □

0.2 Properties of the Integers

◆ Well-Ordering of $\mathbb{Z}$: If $A \subseteq \mathbb{Z}^+$ is nonempty, there exists $a' \in A$ such that $a' \leq a$ for all a in A, called the minimal element of A.
◆ Divides: For $a, b \in \mathbb{Z}$ with $a \neq 0$, we say that a divides b, denoted $a \mid b$, if there exists $c \in \mathbb{Z}$ such that $b = ac$. Moreover, if a does not divide b, we write $a \nmid b$.
◆ Greatest Common Divisor: Denoted as (a, b) for $a, b \in \mathbb{Z}$ with $a \neq 0$, it is the unique $d \in \mathbb{Z}^+$ such that

- $d \mid a$ and $d \mid b$, and
- if $c \in \mathbb{Z}$ such that $c \mid a$ and $c \mid b$, then $c \mid d$.

◆ Relatively Prime: Nonzero $a, b \in \mathbb{Z}$ such that $(a, b) = 1$.
◆ Least Common Multiple: Denoted as $\operatorname{lcm}(a, b)$ for $a, b \in \mathbb{Z}$ with $a \neq 0$, it is the unique $\ell \in \mathbb{Z}^+$ such that

- $a \mid \ell$ and $b \mid \ell$, and
- if $c \in \mathbb{Z}$ such that $a \mid c$ and $b \mid c$, then $\ell \mid c$.

✍ Note

> For $a, b \in \mathbb{Z}$ with $a \neq 0$, we have
>
> $$(a, b) \cdot \operatorname{lcm}(a, b) = |ab|.$$

◆ The Division Algorithm: For $a, b \in \mathbb{Z}$ with $b > 0$, there exist unique $q, r \in \mathbb{Z}$ such that

$$a = bq + r \quad \text{and} \quad 0 \leq r < b.$$

We say that q is the quotient and r is the remainder of the division of a by b.
◆ The Euclidean Algorithm: For $a, b \in \mathbb{Z}$ with $b > 0$, we can find (a, b) by repeatedly applying the division algorithm to a and b until we reach a remainder of 0. More precisely, we can find (a, b) by finding a sequence of integers $r_0, r_1, \ldots, r_n$ such that

$$\begin{aligned} a &= bq_0 + r_0, \\ b &= r_0 q_1 + r_1, \\ r_0 &= r_1 q_2 + r_2, \\ &\vdots \\ r_{n-2} &= r_{n-1} q_n + r_n, \\ r_{n-1} &= r_n q_{n+1} + 0. \end{aligned}$$

The last nonzero remainder r_n is (a, b).
◆ $\mathbb{Z}$-Linear Combination: For $a, b \in \mathbb{Z}$, an expression of the form $ax + by$ for some $x, y \in \mathbb{Z}$.

✍ Note

> We may reverse the Euclidean algorithm (also known as the extended Euclidean algorithm) to obtain the $\mathbb{Z}$-linear combination of a, b that equates to (a, b).

✓ Example ➤ Calculating a Linear Combination

Let $a = 63$ and $b = 39$. Using the Euclidean algorithm, we find

$$\begin{aligned} 63 &= 39(1) + 24, \\ 39 &= 24(1) + 15, \\ 24 &= 15(1) + 9, \\ 15 &= 9(1) + 6, \\ 9 &= 6(1) + 3, \\ 6 &= 3(2) + 0. \end{aligned}$$

Thus, $(63, 39) = 3$. Reversing the Euclidean algorithm, we find

$$\begin{aligned} 3 &= 9 - 6(1) = 9 - (15 - 9(1))(1) \\ &= 9(2) - 15(1) = (24 - 15(1))(2) - 15(1) \\ &= 24(2) - 15(3) = 24(2) - (39 - 24(1))(3) \\ &= 24(5) - 39(3) = (63 - 39(1))(5) - 39(3) \\ &= 63(5) - 39(8). \end{aligned}$$

◆ Prime: A nonzero integer p whose only positive divisors are 1 and p.

◆ Composite: A nonzero integer n that is not prime.

✍ Note

If a prime $p \mid ab$ for some $a, b \in \mathbb{Z}$, then either $p \mid a$ or $p \mid b$.

◆ Fundamental Theorem of Arithmetic: Every integer $n > 1$ can be written as a product of primes, and this factorization is unique up to the order of the factors.

✍ Note

If n has two prime factorizations

$$n = \prod_{i=1}^{k} p_i^{\alpha_i} = \prod_{j=1}^{m} q_j^{\beta_j}$$

for primes p_i, q_j and $\alpha_i, \beta_j \in \mathbb{Z}^+$, then $k = m$ and, after reordering, $p_i = q_i$ and $\alpha_i = \beta_i$ for all i.

✍ Note

Let $a, b \in \mathbb{Z}^+$ have prime factorizations

$$a = \prod_{i=1}^{k} p_i^{\alpha_i}, \quad b = \prod_{i=1}^{k} p_i^{\beta_i}$$

where p_i are distinct primes and $\alpha_i, \beta_i \geq 0$. Then

$$(a, b) = \prod_{i=1}^{k} p_i^{\min(\alpha_i, \beta_i)}, \quad \operatorname{lcm}(a, b) = \prod_{i=1}^{k} p_i^{\max(\alpha_i, \beta_i)}.$$

◆ Euler φ-Function: For $n \in \mathbb{Z}^+$, $\varphi(n)$ is the number of integers k such that $1 \leq k \leq n$ and $(k, n) = 1$, along with two properties:

- For prime p and $\alpha \in \mathbb{Z}^+$, we have $\varphi(p^\alpha) = p^\alpha - p^{\alpha-1}$.
- For $m, n \in \mathbb{Z}^+$ such that $(m, n) = 1$, we have $\varphi(mn) = \varphi(m)\varphi(n)$.

0.3 $\mathbb{Z}/n\mathbb{Z}$: The Integers Modulo n

✍ Note

Let $n \in \mathbb{Z}^+$. Define the relation $\sim$ on $\mathbb{Z}$ by

$$a \sim b \iff n \mid (a - b).$$

- Since $a - a = 0$ and $n \mid 0$, we have $a \sim a$ for all $a \in \mathbb{Z}$.
- If $a \sim b$, then $n \mid (a - b)$, so $nk = a - b$ for some $k \in \mathbb{Z}$. Then $b - a = -nk$, so $n \mid (b - a)$, and thus $b \sim a$.
- If $a \sim b$ and $b \sim c$, then $n \mid (a - b)$ and $n \mid (b - c)$, so $nk_1 = a - b$ and $nk_2 = b - c$ for some $k_1, k_2 \in \mathbb{Z}$. Then

$$a - c = (a - b) + (b - c) = nk_1 + nk_2 = n(k_1 + k_2),$$

so $n \mid (a - c)$, and thus $a \sim c$.

◆ Congruent Modulo n: If $a \sim b$, we write $a \equiv b \pmod{n}$.
◆ Congruence Class: For $a \in \mathbb{Z}$, the congruence class of a modulo n is

$$\begin{aligned}\bar{a} &= \{b \in \mathbb{Z} \mid b \equiv a \pmod{n}\} \\ &= \{a + nk \mid k \in \mathbb{Z}\} \\ &= \{a, a \pm n, a \pm 2n, a \pm 3n, \dots\}.\end{aligned}$$

◆ Integers Modulo n: The set of all congruence classes modulo n is denoted by $\mathbb{Z}/n\mathbb{Z}$. There are precisely n distinct congruence classes modulo n, which can be represented by the integers $0, 1, 2, \dots, n-1$. Thus,

$$\mathbb{Z}/n\mathbb{Z} = \{\bar{0}, \bar{1}, \dots, \overline{n-1}\}.$$

◆ Modular Arithmetic: For any $\bar{a}, \bar{b} \in \mathbb{Z}/n\mathbb{Z}$, we define

$$\bar{a} + \bar{b} = \overline{a + b} \quad \text{and} \quad \bar{a} \cdot \bar{b} = \overline{ab},$$

i.e., we add and multiply congruence classes by adding and multiplying their representatives, where any representative can be chosen.

Theorem 0.3 ➤ Modular Arithmetic is Well-Defined

The operations of addition and multiplication on $\mathbb{Z}/n\mathbb{Z}$ are well-defined, i.e., they do not depend on the choice of representatives. In other words, if $s_1, s_2 \in \mathbb{Z}$ and $t_1, t_2 \in \mathbb{Z}$ such that $\overline{s_1} = \overline{s_2}$ and $\overline{t_1} = \overline{t_2}$, then $\overline{s_1 + t_1} = \overline{s_2 + t_2}$ and $\overline{s_1 t_1} = \overline{s_2 t_2}$. More precisely, if

$$s_1 \equiv s_2 \pmod{n} \quad \text{and} \quad t_1 \equiv t_2 \pmod{n},$$

then

$$s_1 + t_1 \equiv s_2 + t_2 \pmod{n} \quad \text{and} \quad s_1 t_1 \equiv s_2 t_2 \pmod{n}.$$

Proof. Suppose $s_1 \equiv s_2 \pmod{n}$ and $t_1 \equiv t_2 \pmod{n}$. Then $n \mid (s_1 - s_2)$ and $n \mid (t_1 - t_2)$, so there exist integers k_1 and k_2 such that

$$s_1 - s_2 = nk_1 \quad \text{and} \quad t_1 - t_2 = nk_2.$$

Then

$$(s_1 + t_1) - (s_2 + t_2) = (s_1 - s_2) + (t_1 - t_2) = nk_1 + nk_2 = n(k_1 + k_2),$$

so $s_1 + t_1 \equiv s_2 + t_2 \pmod{n}$.

Similarly,

$$\begin{aligned} s_1t_1 - s_2t_2 &= s_1t_1 - s_1t_2 + s_1t_2 - s_2t_2 \\ &= s_1(t_1 - t_2) + (s_1 - s_2)t_2 \\ &= s_1(nk_2) + (nk_1)t_2 \\ &= n(s_1k_2 + k_1t_2), \end{aligned}$$

thus $s_1t_1 \equiv s_2t_2 \pmod{n}$. □

◆ $(\mathbb{Z}/n\mathbb{Z})^\times$: The subset of $\mathbb{Z}/n\mathbb{Z}$ consisting of all congruence classes with multiplicative inverses, i.e.,

$$(\mathbb{Z}/n\mathbb{Z})^\times = \{\bar{a} \in \mathbb{Z}/n\mathbb{Z} \mid \text{there exists } \bar{b} \in \mathbb{Z}/n\mathbb{Z} \text{ such that } \bar{a} \cdot \bar{b} = \bar{1}\}.$$

♣ **Proposition 0.4** ➤ $(\mathbb{Z}/n\mathbb{Z})^\times = \{\bar{a} \in \mathbb{Z}/n\mathbb{Z} \mid (a, n) = 1\}$

See Exercises 0.3.12 and 0.3.13.

1 Introduction to Groups

1.1 Basic Axioms and Examples

◆ Binary Operation: On a set G, it is the function $\star : G \times G \to G$. We write $a \star b$ for $\star(a, b)$, where $a, b \in G$.
◆ Associative: A binary operation $\star$ on a set G such that $a \star (b \star c) = (a \star b) \star c$ for all $a, b, c \in G$.
◆ Commutative: A binary operation $\star$ on a set G such that $a \star b = b \star a$ for all $a, b \in G$. If $a \star b = b \star a$ for *some* $a, b \in G$, we say that a and b commute.

✓ Example ➤ Binary Operations

(1) $+$ and $\cdot$ are associative and commutative binary operations on $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$.
(2) $-$ is not a binary operation on $\mathbb{Z}^+$ since $a - b \notin \mathbb{Z}^+$ when $a < b$, i.e., $- : \mathbb{Z}^+ \times \mathbb{Z}^+ \to \mathbb{Z}^+$ is not well-defined.
(3) The operation of taking cross products in $\mathbb{R}^3$ is a non-associative and non-commutative binary operation on $\mathbb{R}^3$.

◆ Closure: Given a binary operation $\star$ on a set G with subset $H \subseteq G$, we say that H is closed under $\star$ if $a \star b \in H$ for all $a, b \in H$.
◆ Group: An ordered pair $(G, \star)$ for a set G and a binary operation $\star$ on G such that
(1) $\star$ is associative, i.e., $a \star (b \star c) = (a \star b) \star c$ for all $a, b, c \in G$,
(2) there exists an identity element $e \in G$ such that $e \star a = a \star e = a$ for all $a \in G$, and
(3) for each $a \in G$, there exists $a^{-1} \in G$, called the inverse of a, such that $a \star a^{-1} = a^{-1} \star a = e$.
◆ Abelian: A group $(G, \star)$ is Abelian if $\star$ is commutative.
◆ Finite Group: A group $(G, \star)$ is finite if G is a finite set.

✓ Example ➤ Groups

(1) $(G, +)$ is an Abelian group for $G = \mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ with $e = 0$ and $a^{-1} = -a$ for all $a \in G$.
(2) The axioms of a vector space V imply that $(V, +)$ is an Abelian group for all $n \in \mathbb{Z}^+$.

◆ Direct Product: Given groups $(G, \star)$ and $(H, \diamond)$, their direct product is the group $(G \times H, *)$ where

$$G \times H = \{(g, h) \mid g \in G, h \in H\}$$

and $*$ is defined for all $(g_1, h_1), (g_2, h_2) \in G \times H$ as

$$(g_1, h_1) * (g_2, h_2) = (g_1 \star g_2, h_1 \diamond h_2).$$

♣ Proposition 1.1 ➤ Group Properties

Let $(G, \star)$ be a group.
(1) The identity element $e \in G$ is unique.
(2) For each $g \in G$, the inverse $g^{-1} \in G$ is unique.
(3) $(g^{-1})^{-1} = g$ for all $g \in G$.
(4) For all $g, h \in G$, $(g \star h)^{-1} = h^{-1} \star g^{-1}$.
(5) For any $g_1, \ldots, g_n \in G$, the value of $g_1 \star g_2 \star \cdots \star g_n$ is independent of how the product is parenthesized, also known as the generalized associative law.

Proof.
(1) Let $e_1, e_2 \in G$ be identity elements. Then

$$e_1 = e_1 \star e_2 = e_2.$$

(2) Let $g \in G$ and let $g_1^{-1}, g_2^{-1} \in G$ be inverses of g. Then

$$g_1^{-1} = g_1^{-1} \star e = g_1^{-1} \star (g \star g_2^{-1}) = (g_1^{-1} \star g) \star g_2^{-1} = e \star g_2^{-1} = g_2^{-1}.$$

(3) Note that $g^{-1}(g^{-1})^{-1} = e = g^{-1}g$. By the uniqueness of inverses, we have $(g^{-1})^{-1} = g$.

(4) Let $k = (g \star h)^{-1}$. Then

$$(g \star h) \star k = e.$$

Left multiply both sides by g^{-1} and use the associativity of $\star$ to obtain

$$\begin{aligned} g^{-1} \star ((g \star h) \star k) &= g^{-1} \star e \\ (g^{-1} \star g) \star (h \star k) &= g^{-1} \\ e \star (h \star k) &= g^{-1} \\ h \star k &= g^{-1}. \end{aligned}$$

Left multiply both sides by h^{-1} and simplify similarly to obtain

$$\begin{aligned} h^{-1} \star (h \star k) &= h^{-1} \star g^{-1} \\ (h^{-1} \star h) \star k &= h^{-1} \star g^{-1} \\ e \star k &= h^{-1} \star g^{-1} \\ k &= h^{-1} \star g^{-1}. \end{aligned}$$

(5) We proceed by induction on n. The base case $n = 1$ is trivial. For cases $n = 2$ and $n = 3$, the result follows from the associativity of $\star$. Suppose for all $k < n$ that the bracketing of a product of the k elements $h_1 \star \cdots \star h_k$ can be reduced to the form

$$h_1 \star (h_2 \star (\cdots \star (h_{k-1} \star h_k)) \cdots).$$

For notation, write the product above as $R(h_1, \ldots, h_k)$. Consider a product of n elements $g_1 \star \cdots \star g_n$ with some arbitrary bracketing. Then we can write the product as

$$\underbrace{(g_1 \star \cdots \star g_i)}_{i \text{ factors}} \star \underbrace{(g_{i+1} \star \cdots \star g_n)}_{n-i \text{ factors}}$$

for some $1 \le i < n$. By the inductive hypothesis, we may write the i factors as $R(g_1, \ldots, g_i)$ and the $n - i$ factors as $R(g_{i+1}, \ldots, g_n)$. Thus, we have

$$g_1 \star \cdots \star g_n = R(g_1, \ldots, g_i) \star R(g_{i+1}, \ldots, g_n).$$

We first show that

$$R(g_1, \ldots, g_i) \star x = R(g_1, \ldots, g_i, x) \text{ for any } x \in G.$$

We proceed by induction on i. The base case $i = 1$ is trivial. For $i = 2$, we have

$$R(g_1, g_2) \star x = (g_1 \star g_2) \star x = g_1 \star (g_2 \star x) = R(g_1, g_2, x) \text{ by the associativity of } \star.$$

Suppose for all $j < i$ that $R(g_1, \ldots, g_j) \star x = R(g_1, \ldots, g_j, x)$ for any $x \in G$. Noting that

$R(g_1, \ldots, g_i) = g_1 \star R(g_2, \ldots, g_i)$, we have

$$\begin{aligned} R(g_1, \ldots, g_i) \star x &= (g_1 \star R(g_2, \ldots, g_i)) \star x \\ &= g_1 \star (R(g_2, \ldots, g_i) \star x) \\ &= g_1 \star R(g_2, \ldots, g_i, x) \\ &= R(g_1, \ldots, g_i, x), \end{aligned}$$

where the second equality follows from the associativity of $\star$ and the third equality follows from the inductive hypothesis.

Using the above claim and putting $x = R(g_{i+1}, \ldots, g_n)$, we have

$$\begin{aligned} g_1 \star \cdots \star g_n &= R(g_1, \ldots, g_i) \star R(g_{i+1}, \ldots, g_n) \\ &= R(g_1, \ldots, g_i, R(g_{i+1}, \ldots, g_n)) \\ &= R(g_1, \ldots, g_n), \end{aligned}$$

where the last equality follows from the definition of R. Thus, by induction, the value of $g_1 \star \cdots \star g_n$ is independent of how the product is parenthesized. □

✍ Note

We may drop the $\star$ symbol and write gh for $g \star h$ when the binary operation is clear from context. Moreover, we may write g^n for the product of n copies of $g \in G$, where $g^0 = e$ and $g^{-n} = (g^{-1})^n$ for all $n \in \mathbb{Z}^+$. Additionally, we may write 1 for the identity element $e \in G$.

♣ **Proposition 1.2 ➤ Left and Right Cancellation Laws**

Let G be a group with $g, h \in G$. Then the equations $gs = h$ and $tg = h$ have unique solutions $s, t \in G$, respectively. In particular,

(1) if $gx = gy$, then $x = y$, and
(2) if $xh = yh$, then $x = y$.

Proof. The equation $gs = h$ can be solved by left multiplying both sides by g^{-1} to obtain $s = g^{-1}h$, which is unique by the uniqueness of inverses. Similarly, $tg = h$ can be solved to obtain the unique solution $t = hg^{-1}$.

The first cancellation law follows from left multiplying both sides of $gx = gy$ by g^{-1} to obtain $x = y$, and the second cancellation law follows from right multiplying both sides of $xh = yh$ by h^{-1} to obtain $x = y$. □

◆ Order: The smallest $n \in \mathbb{Z}^+$ for a group element $g \in G$ such that $g^n = 1$, denoted as $|g| = n$. If no such n exists, we say that g has infinite order and write $|g| = \infty$.
◆ Multiplication/Group Table: Put $G = \{g_1, g_2, \ldots, g_n\}$ with $g_1 = 1$. Then the multiplication table or group table of G is the $n \times n$ matrix whose (i, j)th entry is $g_i g_j$ for all $1 \le i, j \le n$.